DERMATOLOGY

Fungal and Parasitic Skin infections

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TUCOM-CA

BUG list for Skin Infections		
Bacteria	**Fungi**	**Parasites**
Staphylococcus aureus	*Malassezia furfur*	*Leishmania tropica*
Streptococcus pyogenes	*Candida spp*	*Creeping eruption and Cutaneous Larva Migrans*
Propionibacterium acnes	*Dermatophytes:Trichophyton Microsporum, Epidermophyton*	*Filarial worms - Wuchereria and Brugiya*
Pseudomonas aeruginosa	*Sporothrix schenckii*	*Dracunculus medinensis*
Bacillus anthracis	*Dematiaceous fungi*	*Sarcoptes scabeii*
Mycobacterium leprae	*Blastomyces dermatitidis*	*Loa loa, Oncocerca volvulus*
Atypical mycobacteria (M.marinum, M.ulcerans, M.chelonae, M.fortuitum, M.scrofulaceum)	*Coccidioidomyces immitis*	*Acanthamoeba* Eye infections
Actinomyces and Nocardia		
Borrelia burgdorferi		
Rickettsia rickettsii		
Neisseria meningitidis		
Leptospira interrogans		

LEARNING & TESTABLE OBJECTIVES

For fungal and parasitic infections of the skin,

1. Describe the various fungi and parasites causing infections of the skin, eyes and lymphatics
2. Describe the morphological characteristics
3. Correlate the epidemiology and transmission and life cycle (in parasites)
4. Associate the disease with its features and complications
5. Select the appropriate tests to make a lab diagnosis
6. Choose the appropriate measures for treatment and prevention

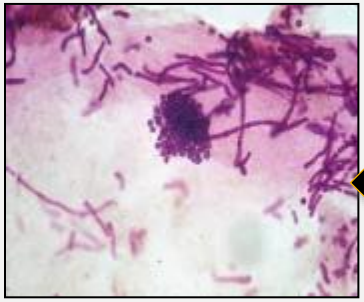
FUNGAL INFECTIONS

Type of lesion	Type of infection	Causative agent (fungi)
Macular scaly lesions	Superficial mycoses-Tinea versicolor	Malassezia furfur/ Pityrosporum orbiculare
Scaly lesions	Cutaneous mycoses • Dermatophytosis (different types of 'Tinea' infections • Candidiasis	Dermatophytes (Trichophyton, Microsporum and Epidermophyton species) Candida species
Nodulo-ulcerative infections	Subcutaneous mycoses and deep mycoses	Sporotrichosis (Sporothrix schenckii) Blastomycosis (Blastomyces dermatitidis) Coccidioidomycoses (Coccidioides immitis)

Superficial mycoses-Tinea versicolor

- Common infection worldwide (more in tropics)
- Superficial infection of the keratinized outer layers of the skin (Stratum corneum), hair and nails.
- Non destructive lesions eliciting little or no host immune response.
- C/F: Small hypo-pigmented/ hyper-pigmented macules on trunk/ extremities/neck
- Well demarcated patches of discoloration/ fine scaling- **NO Pain/ Itch**

- Causative agent: : *Malassezia furfur* also k/a *Pityrosporum orbiculare*, a lipophilic yeast.
- Mode of infection: direct / indirect transfer of keratinized material from person to person.



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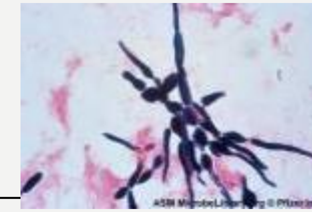
Lab diagnosis:
•Wood's lamp examination of lesion shows yellowish/ orangish fluorescence under UV light.
•Microscopy: Skin scrapings visualized in 10% KOH.
Microscopy: Spaghetti and meat ball appearance of Cigar butt like hyphae and meat ball like spores
•Culture: SDA with olive oil overlay

Treatment: Topical azoles ,selenium sulfide shampoo

Cutaneous mycoses- Candidiasis

Causative agent: Yeast like fungus- *Candida albicans* and other species



- Pruritic rash with erythematous scaly lesions. Common in moist areas of the skin (groin, axilla, toe webs, breast folds).
- Onychomycosis and paronychia (nail involvement)
- Diaper rash

Chronic muco-cutaneous candidiasis is associated with T lymphocyte deficiency.

Lab Diagnosis-

Skin/ Mucosal scrapings:

•Microscopy in 10-20% KOH : Budding Yeast like cells and pseudohyphae.

•Gram stain- gram positive yeast cells and pseudohyphae

Culture:

•Sabouraud's dextrose agar shows creamy, pasty colonies.

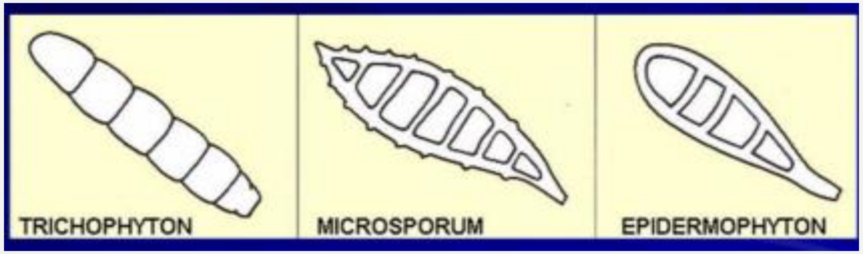
•Germ tube test for identification of Candida albicans (incubation of yeast cells with human serum at 37deg C results in the formation of germ tubes)

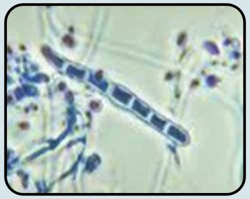
Treatment: Topical imidazoles like Miconazole/ Clotrimazole

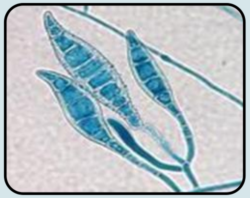
Cutaneous mycoses- Dermatophytosis

- Cutaneous condition caused by a group of filamentous fungi that are keratinophilic and keratinolytic .
- Invade skin (Stratum corneum), hair and nails.
- Dermatophytosis is also called Tinea(ring worm infection)
- 3 genera- Trichophyton, Microsporum and Epidermophyton

A. Zoophilic: associated with animals. Eg: *Microsporum canis, T.verrucosum*.
B. Anthropophilic: associated with humans. Eg: *M.audouinii, T.rubrum*.
C. Geophilic : associated with soil. Eg. *M.gypseum*







Trichophyton

Pencil shaped or long cigar shaped macroconidia and tear drop shaped microconidia

Infects skin, hair and nails

Microsporum

Large, multicellular fusiform macroconidia and few microconidia

Infects skin and hair

Epidermophyton

Clusters of two or three macroconidia

Infects skin and nails

Tinea infections-different presentations

Erythematous, scaly, ring like lesions with advancing border.
Itching, redness +

Tinea corporis

Tinea capitis

Tinea manuum

Tinea pedis

Tinea barbae

Favus: very severe scalp infection with hair loss

Tinea cruris

Tinea unguium

Lab diagnosis and Treatment

- Wood's lamp exam: Bright yellow-green fluorescence with Microsporum
- Nail scrapings/ skin scrapings/ hair clippings: KOH mount (10-20%) show hyphae and conidia. Filamentous, hyaline hyphal elements are seen.
- Culture on Sabouraud Dextrose Agar and dermatophyte test media used for identification.
- Further species identification based on features like the presence of and shape of macroconidia and microconidia.

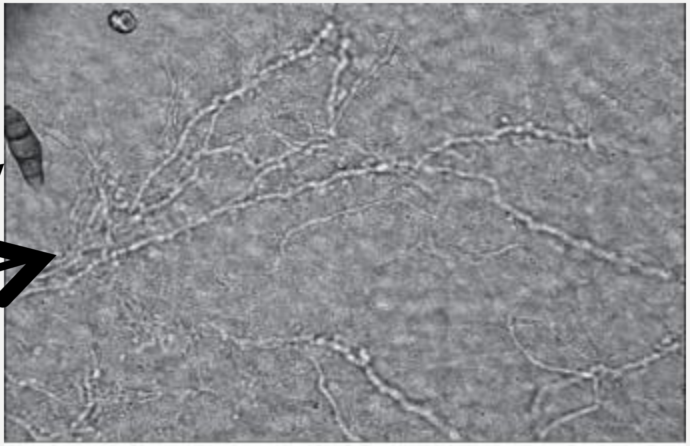


Figure 1: Skin scraping and KOH mount showing branching fungal hyphae in dermatophyte infection

Treatment:

- Topical azoles such as Clotrimazole and Miconazole
- Oral antifungals only in extensive infections: Griseofulvin, fluconazole.

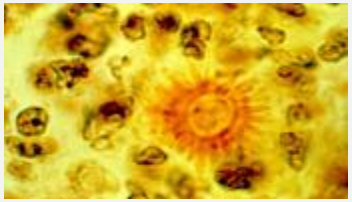
SUB CUTANEOUS MYCOSES
Sporotrichosis/ Rose Gardener's disease

Endemic in sub Himalayan region in Asia; Mexico, Brazil, Peru, Colombia
People at risk-Rose gardeners, Farmers, Florists, Basket weavers

Mode of infection: Traumatic inoculation of soil or vegetable/ organic matter contaminated with the fungus

A subcutaneous infection characterized by a **small nodule that ulcerates with the appearance of a linear chain of secondary lymphatic nodules, two weeks after the appearance of the primary lesion.** These may also ulcerate with pus discharge.

Causative agent, *Sporothrix schenckii* is a dimorphic fungus with yeast forms at 37degc (tissue form) and mold form at colder temperatures (environmental form).



Asteroid bodies seen in lesion biopsy

Microscopic picture of yeast form: **cigar shaped yeasts**

Microscopic picture of mold form: **Hyphae with rosettes/ daisy petal arrangement of abundant oval conidia on conidiophores.**

Treatment:
Itraconazole/ Oral Potassium iodide

SUB CUTANEOUS MYCOSES
Chromoblastomycosis

- Tropical disease caused by soil fungi
  - *Philaphora*, *Cladophialophora* and *Fonsecaea*
  - Seen in Texas and Louisiana
  - Chronic mycosis via traumatic implantation
- Clinical picture
  - Verrucous (wart-like) lesions on feet or legs
  - Especially in persons who walk barefoot
- Diagnosis
  - Histology
    - Brown, round, sclerotic bodies in tissue
    - Copper-colored septate cells
  - Culture
    - Dark molds (dematiaceous)
- Treatment
  - Surgery and itraconazole

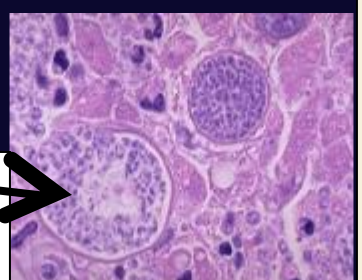
SUB CUTANEOUS MYCOSES
Eumycotic mycetoma

- Tropical disease caused by soil fungi
    - *Madurella* and *Petriellidium* spp.
    - Via trauma, usually to foot
- Clinical picture
    - Swelling with pain, tissue destruction
    - Multiple granulomas and abscesses with drainage
        - Contain darkly colored granules
            - Colonies of infecting organism ➔ inflammation
- Diagnostic microscopy
    - Granules composed of septate hyphae
- Treatment
    - No efficacious antifungal agent; surgery/amputation

Madura foot

DEEP/ SYSTEMIC **MYCOSES-**
Coccidioidomycosis

Skin manifestations- bumps and nodules



Spherules packed with yeasts forms in tissue section

Arthroconidia- mold form

Causative agent-Coccidioides immitis

Endemic regions-Southern California, Arizona, New Mexico, Texas, Nevada

•Mode of infection-Inhalation of **arthroconidia found in desert sand.**

•Within the human host, it converts into spherules containing endospores

•Spread to lungs, skin, reticulo-endolial system etc.

•Associated with Desert Rheumatism/ Desert Bumps; San Joaquin Valley fever, pneumonia.

Lab Diagnosis: Biopsy sample-

- **Microscopy-Giemsa stain shows spherules packed with yeasts forms in tissue section**
- **Culture- Chocolate agar (37degC)-YEAST forms**
  SDA(25degC)- MOLD forms with arthroconidia
- **Serology/ PCR**

(Highly infectious mold- requires high levels of containment while processing in lab)

Treatment- Ketoconazole/ Amphotericin B

DEEP/ SYSTEMIC MYCOSES- Blastomycosis

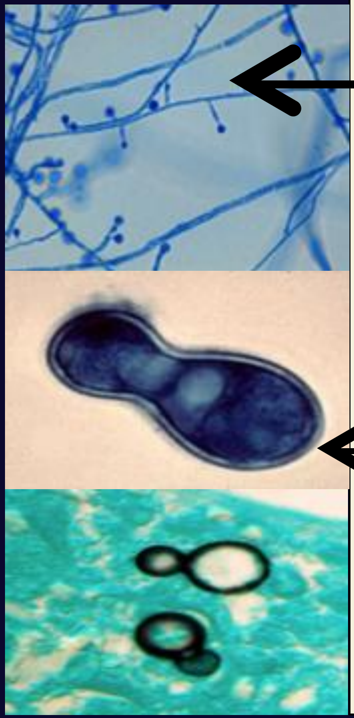
Skin lesion-Erythema nodosum-
Ulcerated verrucous granulomas

Clinical diseases:
Acute/ chronic pulmonary blastomycosis
Disseminated disease
Extrapulmonary: Cutaneous, bone lesions

Causative agent-Blastomyces dermatitidis

Endemic in

- **Ohio, Mississippi river valleys**
- **Northern Minnesota, Canada**
- **North and South Carolina**
- Source: Decaying organic matter such as rotting wood.

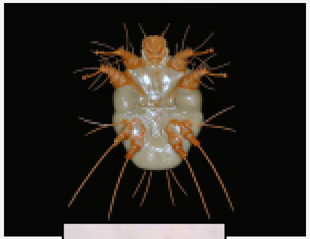


- Environmental form- mold form consisting of hyphae with conidia.
- **Tissue form-characteristic yeast cells with broad based budding and refractile cell walls.**
- Diagnosis: histopathology: demonstration of broad based buds
- Serological tests

Treatment- Ketoconazole/ Amphotericin B

PARASITIC INFECTIONS OF SKIN/ EYE AND LYMPHATIC TISSUE

SCABIES CAUSED BY ITCH MITE *SARCOPTES SCABIEI*



- Eight-legged arthropods with sac like bodies and no antennae.
- They create serpiginous burrows in the upper layers of the epidermis. Eggs are laid and break into larval and nymph stages before becoming adult mites.

Pruritic tracks or papules which are seen on hands, wrists, axillary folds and genitals.
In immuno-compromised host, extensive crusted dermatitis called Norwegian scabies is seen.
Secondary infection such as pyoderma may be seen

- Transmission: contact with person or fomites.
- Overcrowded unhygienic conditions, wartime etc.
- Day care centers, prisons, nursing homes.

- Treatment:

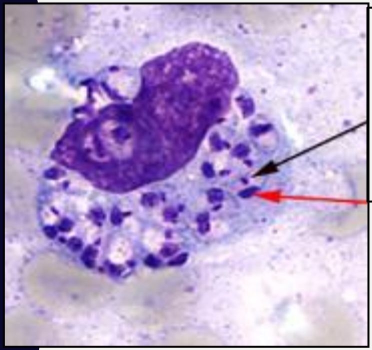
5% Permethrin cream (ELIMITE) application.

Gamma benzene hexachloride (Lindane, used earlier)

CUTANEOUS LEISHMANIASIS

A form of **leishmaniasis** that develops within a few weeks or months of a bite with the Phlebotomous fly (sand fly). Caused by the parasite *L. tropica / L. mexicanal L. (Viannia) brazilensis.*
Begins as papules or nodules and may end up as ulcers with a raised edge.

Endemic regions: **(Afghanistan, Brazil, Iran, Peru, Saudi Arabia and Syria)**
Cutaneous Leishmaniasis has been occurring in American and Canadian troops coming back from Afghanistan, Iraq, Kuwait OR due to visit to tourist destinations like Costa Rica



Lab diagnosis. Giemsa stained smears of Skin biopsy shows Amastigotes (oval, nonflagellated forms) measuring 2-3 um in size. within skin macrophages/clasmatocytes.

Topical application- Paromomycin
Oral agents- Miltefosine
Parenteral agents- Stibogluconate sodium, Amphotericin B

HOOKWORM: GROUND ITCH/ CREEPING ERRUPTION/CUTANEOUS LARVA MIGRANS

- A condition caused by larvae of the nematode- Hookworm.
- Eruptions with serpiginous tracts with severe itching caused due to animal hookworm larvae- *Ancylostoma braziliense or A. caninum* referred to as sandworm as they are found in sandy soil.
- Infecting form: Filariform larvae.
- MOI: Penetration of skin
- Areas: moist soil/ sand etc. that has larvae.
- Treatment: Thiabendazole / Ivermectin

PARASITIC EYE INFECTIONS

- Tissue nematodes, transmitted by flies, in Africa

  Loa loa (African eye worm)/- causes Loiasis

  - Allergic reaction to worms migrating thru subcutaneous tissue = Calabar swelling
  - Adult worms migrate under conjunctiva
  - Transmission by Mango flies of Chrysops genus.

Onchocerca volvulus/Onchocerciasis
Transmitted by Black fly-Simulium
Leading cause of blindness = river blindness
Allergic reaction = subcutaneous nodules
Microfilaria migrating thru eye = blindness

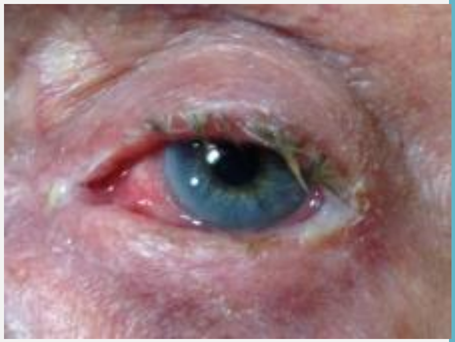
Free-living amoeba

Acanthamoeba

Keratitis in healthy people

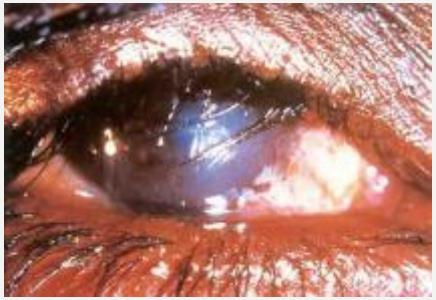
BACTERIAL EYE INFECTIONS-FYI

- *Staphylococcus aureus, Streptococcus pneumoniae, Haemophilus influenzae, Moraxella catarrhalis*
  - Acute purulent conjunctivitis



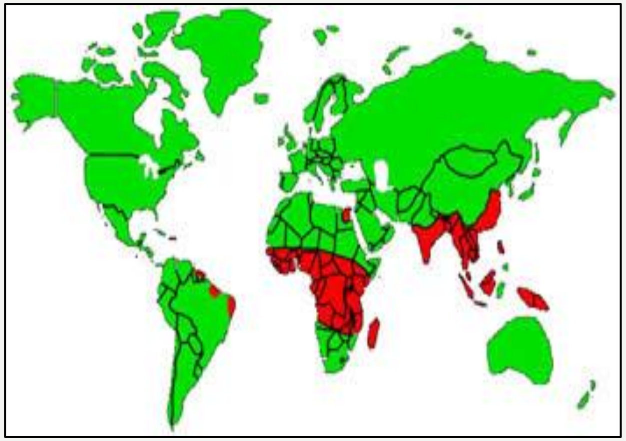
- *Neisseria gonorrhoeae*
  - Gonococcal ophthalmia neonatorum

- *Chlamydia trachomatis* serovars A-C
  - Trachoma
  - Conjunctival and corneal scarring
    - Inturned lashes
  - Leading cause of preventable blindness



FILARIASIS/ ELEPHANTIASIS

- *Wuchereria bancrofti (Bancroftian filariasis)*
- *Brugia malayi (Malayan filariasis)*
  - Tissue nematodes
    - Contain endosymbiotic bacteria *Wolbachia*

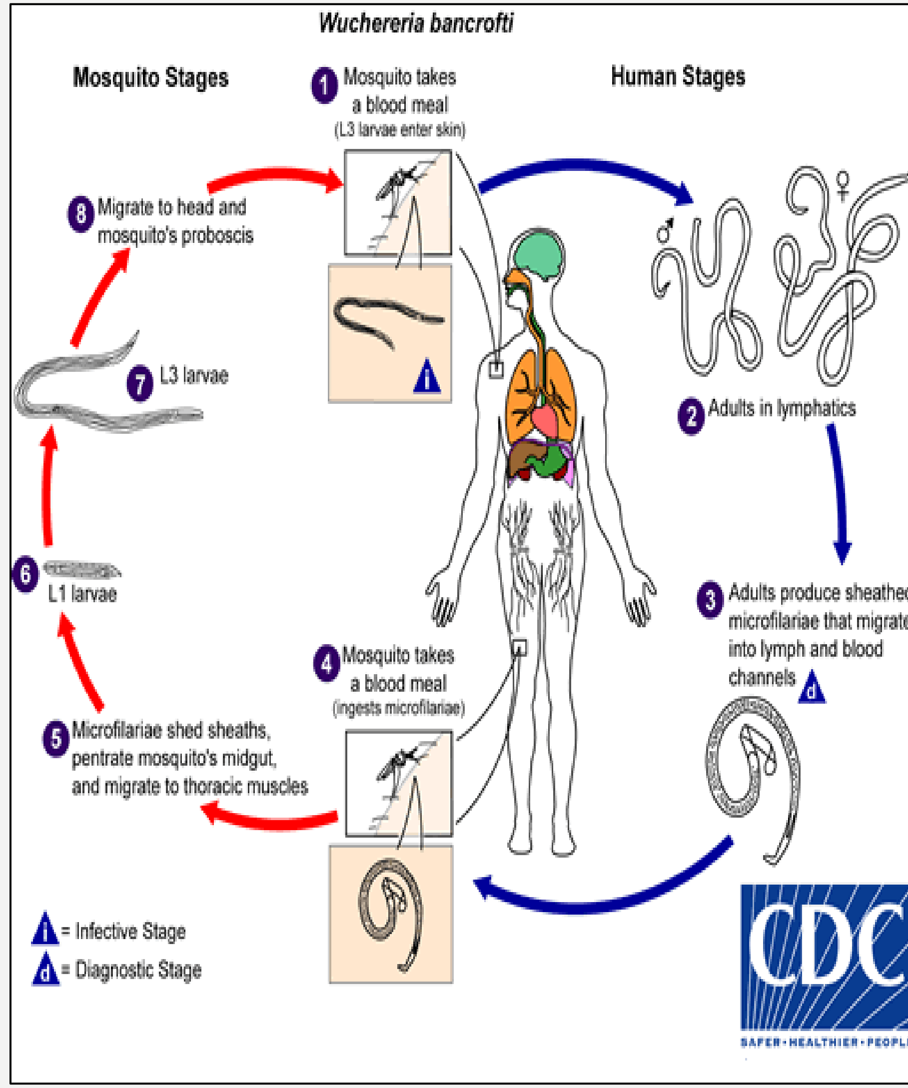


Clinical presentation ~ months to a year after infection
- Fever, eosinophilia,
- Lymphatic disease, lymphedema
- Tissue hypertrophy and elephantiasis

Sub-Saharan Africa, Southeast Asia, the India, China, Malaysia, Philippines, Indonesia, Pacific islands

- Larvae transmitted by mosquito bite (Culex)
  - Larva migrate to lymphatic system
    - Primarily arms, legs, groin
  - Mature into adult forms (seen in lymphatics and lymph nodes). They reproduce ➔ microfilaria enter blood at night
    - Diagnostic stage (seen in peripheral smear)

Microfilaria of *Wuchereria bancrofti* with sheath in the peripheral blood of human (Giemsa stain; original magnification, ×200).



TREATMENT-
Diethycarbamazine
Ivermectin

DRACUNCULIASIS

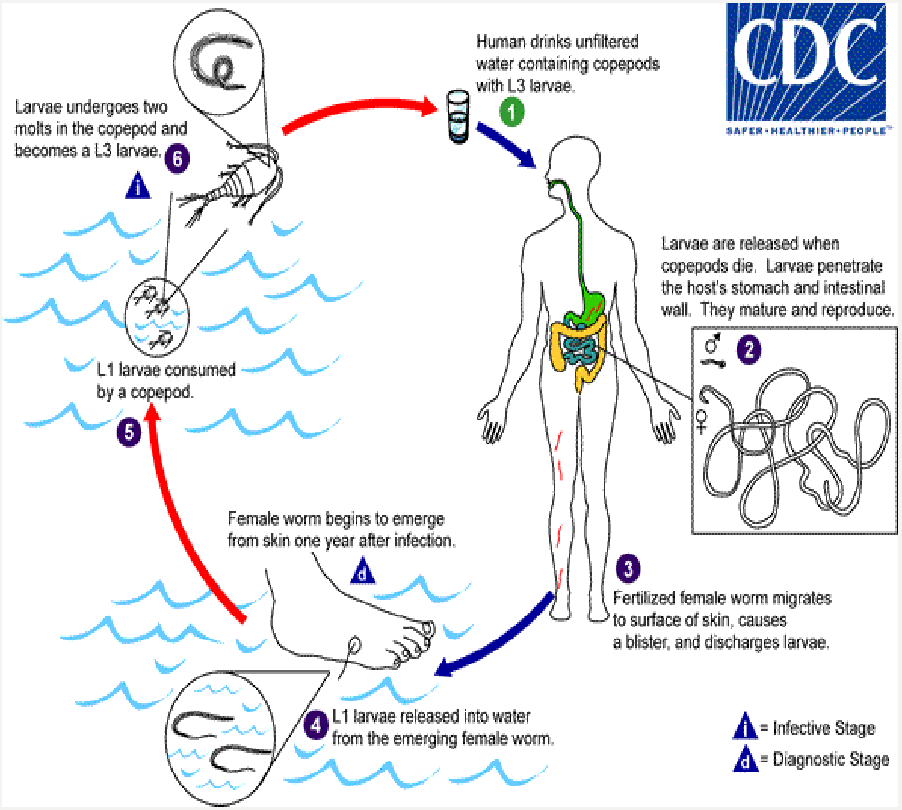
- *Dracunculus medinensis*
  - AKA Guinea worm, tissue nematode
  - Man is the definitive host.
  - Transmission by consumption of water containing infected copepods(cyclops)-the intermediate host
  - Fertilized females migrate to skin ➔papule ➔ release larva (humans are definitive host)

Painful papule

Emerging worm

DRACUNCULUS MEDINENSIS

Lifecycle



Treatment

Consists of slow extraction of the worm combined with wound care and pain management


Thank you!!